Calculator using Java Swing following MVC Architecture

Title

Design and Develop a Calculator using Java Swing following MVC Architecture

Aim

To design a GUI-based calculator using Java Swing that performs basic arithmetic operations using the MVC (Model-View-Controller) architecture.

Objectives

- Understand MVC Architecture.
- Learn separation of Model, View and Controller.
- Practice Swing Components.
- Handle button click events.
- Perform arithmetic operations.

Theory

What is MVC?

MVC stands for

- **Model**
- **View**
- **Controller**

It separates the application into three parts.

1. Model

Contains business logic.
Example

```
Addition
Subtraction
Multiplication
Division
```

No GUI code is written here.

2. View

Contains GUI only.
Components used

- JFrame
- JLabel
- JTextField
- JButton

The View never performs calculations.

3. Controller

Acts as a bridge.
Responsibilities

- Reads data from View.
- Calls Model.
- Displays result back in View.

MVC Diagram

```
+-------------+
|    View     |
|  Swing GUI  |
+-------------+
       |
       |
       V
+-------------+
| Controller  |
|ActionListener|
+-------------+
       |
       |
       V
+-------------+
|    Model    |
|Calculations |
+-------------+
```

Components Used

```
JFrame
JLabel
JTextField
JButton
JPanel
ActionListener
```

Algorithm

1. Create Model class.
2. Create View class.
3. Create Controller class.
4. Enter two numbers.
5. Click any operation button.
6. Controller sends values to Model.
7. Model performs calculation.
8. Result is displayed.

Program

Model.java

```
public class CalculatorModel {
    public double add(double a, double b) {
        return a + b;
    }
    public double subtract(double a, double b) {
        return a - b;
    }
    public double multiply(double a, double b) {
        return a * b;
    }
    public double divide(double a, double b) {
        if (b == 0)
            return 0;
        return a / b;
    }
}
```

View.java

```
import javax.swing.*;
public class CalculatorView extends JFrame {
    JTextField txtNum1, txtNum2, txtResult;
    JButton btnAdd, btnSub, btnMul, btnDiv;
    public CalculatorView() {
        setTitle("MVC Calculator");
        setSize(400,300);
        setLayout(null);
        JLabel l1=new JLabel("First Number");
        l1.setBounds(30,30,100,25);
        add(l1);
        txtNum1=new JTextField();
        txtNum1.setBounds(150,30,150,25);
        add(txtNum1);
        JLabel l2=new JLabel("Second Number");
        l2.setBounds(30,70,100,25);
        add(l2);
        txtNum2=new JTextField();
        txtNum2.setBounds(150,70,150,25);
        add(txtNum2);
        btnAdd=new JButton("+");
        btnAdd.setBounds(30,120,60,30);
        add(btnAdd);
        btnSub=new JButton("-");
        btnSub.setBounds(100,120,60,30);
        add(btnSub);
        btnMul=new JButton("*");
        btnMul.setBounds(170,120,60,30);
        add(btnMul);
        btnDiv=new JButton("/");
        btnDiv.setBounds(240,120,60,30);
        add(btnDiv);
        JLabel l3=new JLabel("Result");
        l3.setBounds(30,180,100,25);
        add(l3);
        txtResult=new JTextField();
        txtResult.setBounds(150,180,150,25);
        txtResult.setEditable(false);
        add(txtResult);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setVisible(true);
    }
}
```

Controller.java

```
import java.awt.event.*;
public class CalculatorController implements ActionListener {
    CalculatorView view;
    CalculatorModel model;
    public CalculatorController(CalculatorView view,
                                CalculatorModel model) {
        this.view=view;
        this.model=model;
        view.btnAdd.addActionListener(this);
        view.btnSub.addActionListener(this);
        view.btnMul.addActionListener(this);
        view.btnDiv.addActionListener(this);
    }
    @Override
    public void actionPerformed(ActionEvent e) {
        double a=Double.parseDouble(view.txtNum1.getText());
        double b=Double.parseDouble(view.txtNum2.getText());
        double result=0;
        if(e.getSource()==view.btnAdd)
            result=model.add(a,b);
        else if(e.getSource()==view.btnSub)
            result=model.subtract(a,b);
        else if(e.getSource()==view.btnMul)
            result=model.multiply(a,b);
        else if(e.getSource()==view.btnDiv)
            result=model.divide(a,b);
        view.txtResult.setText(String.valueOf(result));
    }
}
```

Main.java

```
public class Main {
    public static void main(String[] args) {
        CalculatorView view = new CalculatorView();
        CalculatorModel model = new CalculatorModel();
        new CalculatorController(view, model);
    }
}
```

Output

```
------------------------------
|     MVC Calculator         |
------------------------------
First Number   [ 25        ]
Second Number  [ 10        ]
 [+] [-] [*] [/]
Result         [ 35.0      ]
------------------------------
```

Viva Questions

1. What is MVC Architecture?
2. What are the three components of MVC?
3. Why do we separate Model and View?
4. What is the purpose of ActionListener?
5. What is the difference between JFrame and JPanel?
6. Why is setEditable(false) used?
7. Which package contains Swing classes?
8. What happens when division by zero occurs?
9. What is event-driven programming?
10. What is the advantage of MVC over writing everything in one class?